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Best Estimates of Binary Gas Mixtures Properties for Closed Brayton Cycle Space Applications

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Noble gases and their binary mixtures at pressures from 0.1 to 20 MPa and temperatures up to 1400 K, are being considered as working fluids in terrestrial and space gas-cooled nuclear reactor systems with Closed Brayton Cycle (CBC) engines. At such high pressures, the Chapman-Enskog approach for dilute gases is not accurate. This paper developed correlations for the thermodynamic (density, enthalpy, specific heat capacity and heat capacity ratio) and transport (dynamic viscosity and thermal conductivity) properties of the 5 noble gases (helium, neon, argon, krypton and xenon) and their 10 binary mixtures for temperatures up to 1400 K and pressures up to 20 MPa. Results show that, at temperatures $\geq$ 400 K and pressures $\leq$ 2 MPa, He, Ne, and the He-Xe and He-Kr binary gas mixtures with a molecular weight of 40 gram/mole behave essentially like perfect gases, and the error of neglecting the effect of pressure on their compressibility factor, specific heats and transport properties is very small ($\leq$ 1%).

Nomenclature

B_{ii}, B	Second virial coefficient for a pure gas (mole / m^3)
B_{ij}	Second virial interaction coefficient (mole / m^3), $i \neq j$
C_{iii}, C	Third virial coefficient of pure gas (mole^2 / m^6),
C_{ijj}	Third virial interaction coefficient (mole^2 / m^6), $i \neq j$
$\hat{C}_P$, $\hat{C}_\upsilon$	Specific heat capacity at constant pressure, constant volume (J / mole.K)
f	Higher-order correction factor for transport property
$\hat{H}$	Molar enthalpy (J / mole)
k	Boltzmann constant, k = 1.380662 x 10^{-23} J / K
m	Mass of one gas molecule, M / N_A (kg)
M	Gas molecular weight (kg / mole)
N	Number of components in a gas mixture
N_A	Avogadro number, N_A = 6.022045×10^{23} mol^{-1}
P	Pressure (Pa)
R_g	Perfect gas constant, 8.31441 J / mol.K
T	Temperature (K)
T_{cr}, T_{ii}^*	Critical temperature of pure gas (K)
T_{ij}^*	Interaction temperature (K), $i \neq j$
$\overline{T}^*$	"Pseudo-critical" temperature (K)
$\hat{U}$	Molar internal energy (J / mole)
V_{ii}^*	Molar volume (m^3/mole), $R_g(T_{cr} / P_{cr})_i$

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V_{ij}^*	Interaction molar volume (m^3/mole), $i \neq j$
x_i	Molar fraction of component i
Z	Compressibility factor

Greek

$\Delta\lambda$	Excess conductivity (W / m.K), $\lambda(T,P) - \lambda^o(T)$
$\Delta\mu$	Excess viscosity (Pa.s), $\mu(T,P) - \mu^o(T)$
ε_{ij}	Depth of molecular potential well (J)
θ	Reduced temperature, T / T_{cr}
λ	Thermal conductivity (W / m.K)
$\overline{\lambda}^*$	"Pseudo-critical" conductivity (W / m.K)
μ	Dynamic viscosity (Pa.s)
$\overline{\mu}^*$	"Pseudo-critical" viscosity (Pa.s)
υ	Gas molar volume (m^3 / mole)
π	Reduced pressure, P / P_{cr}
ρ	Massic density (kg / m^3)
$\hat{\rho}$	Molar density, $\rho / M = 1/\upsilon$ (mole / m^3)
ρ_r	Reduced density, ρ / ρ_{cr}
σ_{ij}	Distance for which molecular potential is zero (m)
Ψ	Dimensionless function of reduce density
$\Omega^{(k,l)*}$	Dimensionless collision integral

Subscript / Superscript

$\hat{}$	Molar property (per mole)
$\overline{\ }$	Gas mixture property
*	Reduced, Critical or "pseudo-critical"
CE	Chapman-Enskog approach
cr	Critical
exp	Experimental
o	Dilute-density; reference value at 0.1 MPa
λ	Thermal conductivity
μ	Dynamic viscosity

I. Introduction

Noble gases and binary mixtures are used or being considered as working fluids for terrestrial and space, gas cooled nuclear reactors with Closed Brayton Cycle (CBC) engines. The gas temperatures in these systems vary from 300 K to 1400 K and the system pressures vary from 0.1 to 20 MPa. At these pressures, the theory of diluted gases based on the Chapman-Enskog approach does not apply. Thus, accurate predictions of the thermodynamic and transport properties of such dense gases and their binary mixtures are needed. Helium (He), with the lowest molecular weight, has the best thermal properties of all noble gases, and is the preferred working fluid in terrestrial nuclear power plants. Conversely, the low molecular weight of He increases the aerodynamic loading and hence, the mass and the size of the turbomachinery in space nuclear reactor power systems.[1]

When He is mixed with heavier noble gases, the thermal properties of the resulting binary mixtures are superior to those of pure gases of equal molecular weight. Recently, El-Genk and Tournier[1] have shown that mixing He with the heavier gases of Krypton (Kr) and Xenon (Xe) to molecular weights of < 22 and < 40 increases slightly the heat transfer coefficient beyond that of He and decreases significantly the loading of the turbomachinery. A He-Xe mixture with a molecular weight of 40 (72 mole% He and 28 mole% Xe) has the same heat transfer coefficient as He, at only 10% of the turbomachinery loading. As a result, using such a He-Xe mixture decreases the heat transfer areas in the nuclear reactor, the radiator heat exchanger, and recuperator in space nuclear reactor power systems with CBC engines, as well as the mass and size of the turbomachinery. These results were based on the correlations developed in this paper for the transport properties of dense nobles gases and binary gas mixtures.

This paper developed semi-empirical correlations for accurately predicting the thermodynamic properties (density, enthalpy, specific heat capacity and heat capacity ratio) and transport properties (dynamic viscosity and thermal conductivity) of noble gases (He, Ne, Ar, Kr and Xe) and their binary mixtures, for temperatures up to 1400 K and pressures up to 20 MPa. These temperatures and pressures cover the operating conditions encountered in space nuclear power systems and in terrestrial nuclear power plants with CBC engines. The developed correlations are based on the Chapman-Enskog approach for dilute gases[2,3] after correcting for the effect of high gas density, a comparison with a large compiled experimental database, and using the law of corresponding states. Estimates of the differences in these properties when neglecting the effect of pressure are presented for the pure noble gases, and helium-xenon mixture with a molecular weight of 40 gram/mole (72 mole% helium and 28 mole% xenon).

II. Motivation

The most widely used approach for determining the thermophysical properties of gases uses a perturbation technique to obtain an approximate solution of the Boltzmann's equation[2] for dilute monoatomic gases. This approach, developed before World War I by Chapman in England and independently by Enskog in Sweden,[3] assumes the gases and their mixtures are sufficiently dilute, such that the molecular dimensions are small, compared to the mean free path of the molecules, which, most of the time, move freely and interact only through binary collisions. Nonetheless, the gas density is high enough to neglect molecule-wall collisions. The thermodynamic state of the gas is represented by a virial expansion, with up to two virial coefficients, and the Knudsen effect is neglected, hence the motion of the molecules is described by the classical Newtonian mechanics. The macroscopic behavior of the gas is thus, described with sufficient accuracy, using a single distribution function in the phase space of pairs of molecules[2] and the equilibrium properties are expressed solely in terms of the pair distribution function. The Chapman-Enskog approach developed expressions of the transport coefficients of the binary gas mixtures in terms of the collision integrals, $\Omega_{i,j}^{(s,l)}$, whose values depend on the molecular interaction potential.[3,4]

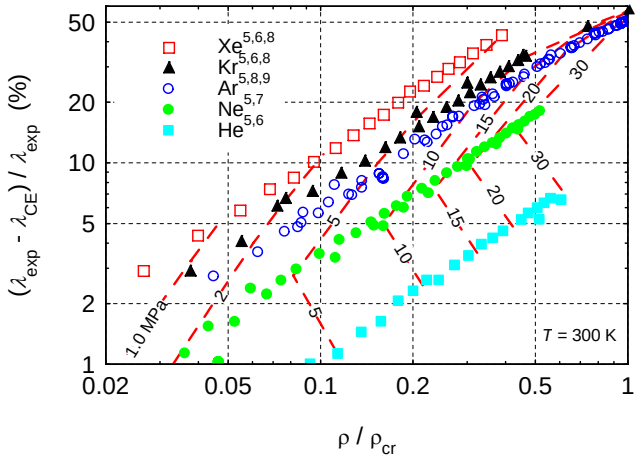


Figure 1. Thermal conductivity of dense noble gases.

Although it accurately predicts the thermophysical properties of dilute gases and binary gas mixtures, the Chapman-Enskog approach is not applicable to dense gases in which multiple molecule collisions occur. To illustrate this point, an extensive database of measured thermal conductivity of noble gases, at 300 K, and up to 30 MPa, is compiled and compared with the calculated values using the Chapman-Enskog approach in Figure 1. As shown in this figure, the difference between the experimental values and the predictions using the Chapman-Enskog approach could be significant. For the same reduced density (ρ/ρ_{cr}), the difference in the thermal conductivity of dense and dilute gases increases with increasing the molecular weight of the gas. The thermal conductivity, as well as the viscosity of dense gases, increases as the pressure and/or the density increase. For example, at a reduced density ρ/ρ_{cr} = 0.2 and 300 K, the thermal conductivity predictions using the Chapman-Enskog approach are lower than measured by 7%, 12%, 16% and 22% for Ne, Ar, Kr and Xe, respectively (Figure 1). For He at the same operating condition, however, the difference is much smaller, ~ 2%. The difference in thermal conductivities of dense and dilute gases increases with increasing pressure and/or molecular weight. For example, this difference for pure helium is 1.2% at 5 MPa and 4.8% at 20 MPa. For the heaviest noble gas, Xe at the same pressures, the differences in the thermal conductivities of dense and dilute gases are 40% and ~90% (Figure 1). Although the dynamic viscosity of noble gases is directly related to their thermal conductivity, the differences between the dense and dilute gas values for the viscosity are smaller than for the conductivity.

The property correlations developed in this work show that, at 5 MPa and 770 K, there is a 5% difference in the calculated thermal conductivity of Xe using the Chapman-Enskog approach and < 5% difference at temperatures > 125 K, 200 K, 440 K and 570 K for He, Ne, Ar and Kr, respectively. At lower temperatures and higher pressure than 5.0 MPa, the difference between the calculated thermal conductivities of dilute gases and measured values exceeds 5%. Because of its very low critical temperature (5.2 K), the difference between measured and calculated

thermal conductivity of He (as well as Ne, T_{cr} = 44.6 K), using the Chapman-Enskog approach at typical operating conditions in terrestrial, gas cooled reactors (300–1300 K and 20 MPa), is negligible.

At 2 MPa, the difference between measured and calculated thermal conductivity of Xe, at a typical compressor temperature of 400 K, is 8%. Since the He-Xe gas mixture is the working fluid of choice in space nuclear reactor power systems with CBC engines,[1] it is important to examine the applicability of the Chapman-Enskog approach for dilute gases to calculating the thermophysical properties of Xe, He and the He-Xe mixtures at the operating conditions in space nuclear reactor power systems with CBC engines[1] (up to 1400 K and 2.0 MPa).

The law of corresponding states for dense gases, with additional coefficients obtained based on the compiled experimental data by other investigators, corrects the Chapman-Enskog approach for predicting the thermal conductivity and dynamic viscosity of the dense noble gases and their binary gas mixtures.

III. Law of Corresponding States for Dense Gases

To account for the effect of low temperature and/or high pressure (or density) on the thermophysical properties of noble gases, the two-parameter law of the corresponding states for dense gases is expressed in terms of T_{cr} and ρ_{cr}, instead of the potential parameters ε and σ in the Chapman-Enskog approach for dilute gases. The critical parameters of pure gases are readily available and accurate. The large experimental database in Figures 2 and 3 shows that at $T > T_{cr}$ and $\rho < 2\ \rho_{cr}$, the reduced excess viscosity, $\Delta\mu^*$, and excess conductivity, $\Delta\lambda^*$, of the gases are solely functions of density.[10-14]

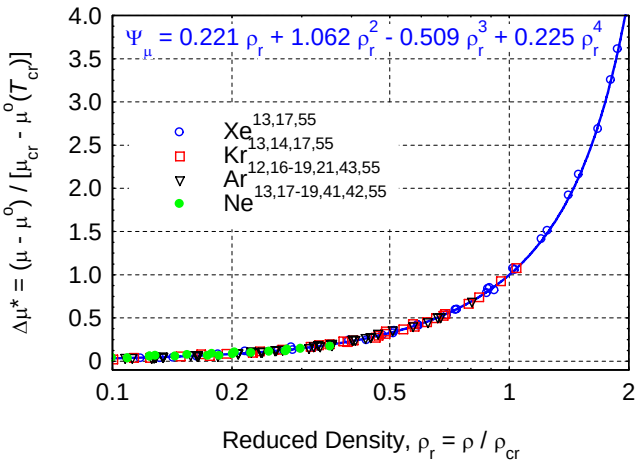


Figure 2. Reduced, excess viscosity of noble gases.

A. Dynamic Viscosity of Pure Gases

The excess viscosity at the critical point, $\Delta\mu_{cr}$, is obtained from the least square fit of the experimental data in Figure 2 at a reduced density of unity. The determined values for Xe, Kr, Ar and Ne are listed in Table 1, which are 4% to 7% higher than those obtained by Hougen and Watson[15] in 1947. The developed viscosity correlation for dense noble gases requires knowledge of the critical viscosity, expressed in terms of the critical parameters, T_{cr} and ρ_{cr}, and a relationship between $\mu^o(T_{cr})$ and μ_{cr}. The values of $\mu^o(T_{cr})$ are obtained from the tables developed by Vargaftik[16] and Rabinovitch et al..[17] A peculiarity of He is that unlike other noble gases, its dynamic viscosity is independent of pressure and solely a function of temperature.[11,18,19] The critical viscosity calculated using the relationship proposed by Trautz is:[3]

$$\mu^*_{cr} = b\sqrt{MT_{cr}}\,/\,V_{cr}^{\,2/3} \quad . \qquad (1)$$

The constant $b = 0.204\times10^{-7}$ is determined from the critical viscosity values of Ne, Ar, Kr and Xe in Figure 2 (Table 1). Equation (1) predicts the critical viscosity of these noble gases within 0.8%, and $\mu_{cr}\,/\,\mu^o(T_{cr})$ ~ 2.30 (Table 1), thus:

$$\mu(T,P) = \mu^o(T) + \left(1 - \frac{1}{2.3}\right)\mu^*_{cr} \times \Psi_\mu\left(\frac{\rho}{\rho_{cr}}\right), \qquad (2)$$

Where Ψ_μ is the unique function shown in Fig. 2. The experimental data[20-22] and tabulated values[4,22,23] of the viscosity of dilute gases at 100 < T < 1400 K are correlated accurately using the expressions:

$$\mu^o(T) = A_\mu\left(T - T_\mu\right)^n \quad . \qquad (3)$$

The values of the coefficients A_μ, T_μ and exponent n are given in Table 1. This expression is in agreement with calculated values for dilute gases[4] to within ± 1.3%.

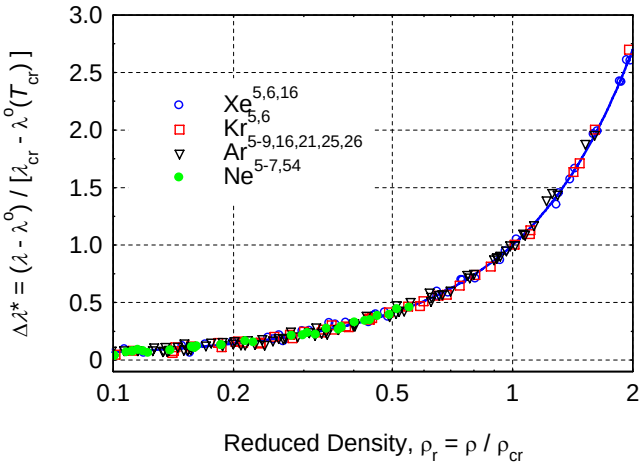


Figure 3. Reduced, excess conductivity of noble gases.

Table 1. Fundamental and critical properties of pure noble gases.

Property	Helium	Neon	Argon	Krypton	Xenon
M (kg/mole)	0.004003	0.020179	0.039948	0.0838	0.13129
T_{cr} (K)	5.2	44.5	150.7	209.5	289.7
P_{cr} (MPa)	0.2275	2.678	4.863	5.51	5.84
ρ_{cr} (kg/m^3)	69.64	481.9	535.6	908.4	1110.
V_{cr} (cm^3/mole)	57.48	41.87	74.59	92.25	118.28
$0.291V^*$ (cm^3/mole)	55.30	40.20	74.98	91.99	120.02
Deviation (%)	-3.781	-3.987	0.526	-0.278	1.473
V^* (cm^3/mole)	190.04	138.16	257.66	316.13	412.45
A_μ x 10^7 (Eq. (3))	3.0629	8.4528	6.9891	6.9629	7.5683
T_μ (K)	-21.33	16.47	65.70	71.07	112.31
n	0.7243	0.642584	0.63977	0.667	0.655473
$\Delta\mu_{cr} \times 10^6$ (Pa.s)	-	8.9	16.0	23.3	29.7
$\mu^o(T_{cr}) \times 10^6$ (Pa.s)	1.428	7.257	12.34	18.39	22.55
$\mu_{cr} \times 10^6$ (Pa.s)	-	16.157	28.34	41.69	52.25
μ_{cr}^* deviation (%)	-	-0.786	-0.329	0.417	-0.076
Ratio $\mu_{cr} / \mu^o(T_{cr})$	-	2.226	2.297	2.267	2.317
$\Delta\mu_{cr}^*$ deviation (%)	-	1.803	-0.215	1.554	-0.639
$\Delta\lambda_{cr} \times 10^3$ (W/m.K)	24.0	24.5	18.62	12.96	10.55
$\lambda^o(T_{cr}) \times 10^3$ (W/m.K)	10.32	10.97	9.603	6.868	5.416
$\lambda_{cr} \times 10^3$ (W/m.K)	34.32	35.47	28.223	19.828	15.966
λ_{cr}^* deviation (%)	6.5	0.47	-0.50	0.99	-0.28
Ratio $\lambda_{cr} / \lambda^o(T_{cr})$	3.326	3.233	2.939	2.887	2.948
$\Delta\lambda_{cr}^*$ deviation (%)	0.490	-4.014	-0.484	1.958	-0.422

B. Thermal Conductivity of Pure Gases

An expression for the thermal conductivity of dense noble gases is developed using the same procedure outlined in the previous section for the dynamic viscosity. When the excess thermal conductivity, $\Delta\lambda$, is plotted versus density, a unique curve is obtained.[24-26] The dilute-gas thermal conductivity, $\lambda^o(T_{cr})$, is obtained from the tables developed by a number of investigators.[6,16,17,27] The empirical relationship of the reduced thermal conductivity, obtained from the least square fit of the experimental data in Figure 3, is:

$$\Delta\lambda^* = 0.645\rho_r + 0.33\rho_r^2 + 0.0368\rho_r^3 - 0.0128\rho_r^4 = \Psi_\lambda(\rho/\rho_{cr}). \quad (4)$$

For the general expression[24] for the critical thermal conductivity of binary mixtures, $\lambda_{cr}^* = a M^c T_{cr}^d (0.291\times V^*)^g$, the values in Table 1 for Ne, Ar, Kr are used to evaluate the coefficient and exponents, thus it becomes:

$$\lambda_{cr}^* = \left(\left(0.304\times10^{-4}\right)T_{cr}^{0.277}\right)/\left(M^{0.465}(0.291\times V^*)^{0.415}\right). \quad (5)$$

As shown in Table 1, the ratio $\lambda_{cr}/\lambda^o(T_{cr})$ for the heavy gases of Ar, Kr and Xe is nearly constant ~ 2.94. The values of $\lambda_{cr}/\lambda^o(T_{cr})$ for He (3.326) and Ne (3.233) are slightly higher; however, since the thermal conductivities of He and Ne exhibit weak dependence on pressure, the error in using 2.94 is inconsequential. Thus, the thermal conductivity of noble gases may be calculated using the following expression:

$$\lambda(T,P) = \lambda^o(T) + \left(1 - \frac{1}{2.94}\right)\lambda_{cr}^* \Psi_\lambda\left(\frac{\rho}{\rho_{cr}}\right). \quad (6)$$

The thermal conductivity of the dilute gases is calculated accurately, in terms of the dynamic viscosity, using the kinetic-theory relationship:[3]

$$\lambda^o = \frac{15}{4}\frac{R_g}{M}\mu^o = \frac{15}{4}\frac{k}{m}\mu^o . \qquad (7)$$

Equation (7) agrees with tabulated [6,23] (100 K < T < 1400 K) and calculated values for dilute gases[4] to within ±2.6%.

C. Density of Pure Gases

The densities of pure gases are calculated using the virial state equation, in which the compressibility factor Z is assumed a second order polynomial of the molar density, thus:

$$P = \rho R_g T \; Z/M = R_g T\left[\hat{\rho} + B\,\hat{\rho}^2 + C\,\hat{\rho}^3\right]. \qquad (8)$$

This approximation is accurate at least up to the critical density.[28] Equation (8), which is easily inverted[29] to give the gas density, predicts the compiled density data and recommended values[30] for noble gases to within less than ± 1%, for $T > T_{cr}$ + 40 K, P < 20 MPa and $\rho < \rho_{cr}$. The virial coefficients in Eq. (8), which are related to the intermolecular potentials,[2] have been experimentally evaluated for the pure noble gases. Figures 4 and 5 show that the normalized virial coefficients for Xe, Kr, Ar and Ne collapse on single curves, confirming the applicability of the principle of corresponding states. The correlation developed by Pitzer and Curl[31] for the second virial coefficient, B, is in excellent agreement with the experimental data when 0.7 < θ < 3.0, but for Ar it deviates from the data when 0.5 < θ < 0.7 and 3.0 < θ < 6.6 and for Ne when 3.0 < θ < 22. The following expression for the second virial coefficient of noble gases, except He, developed based on experimental data by a number of investigators (Figure 4), is accurate over the range of 0.5 < θ < 22:

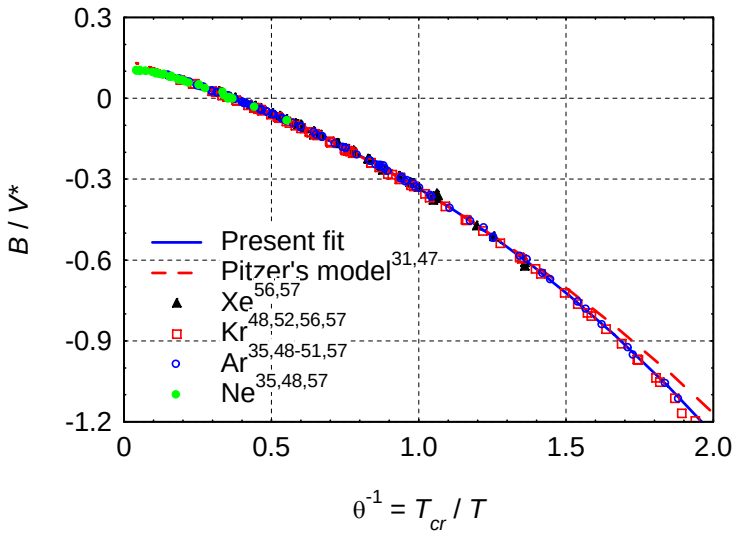


Figure 4. Reduced 2nd virial coefficients of noble gases.

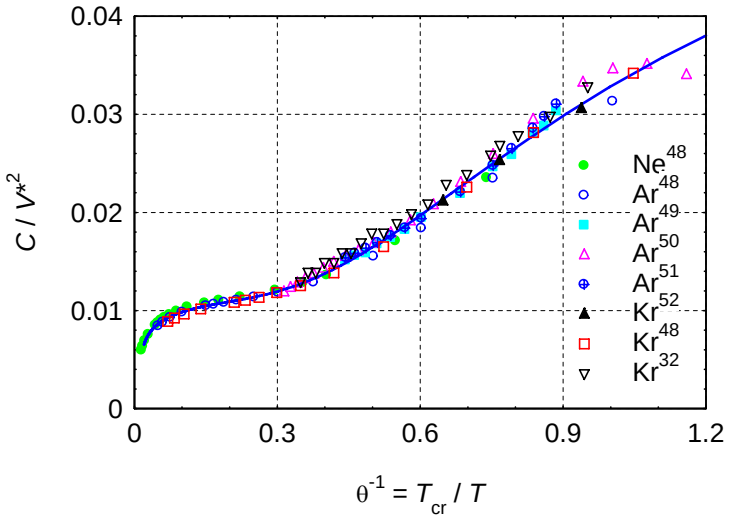


Figure 5. Reduced 3rd virial coefficients of noble gases.

$$\left(B/V^*\right) = -102.6 + \left(102.732 - 0.001\times\theta - \frac{0.44}{\theta^{1.22}}\right)\times\tanh\left(4.5\sqrt{\theta}\right) = \Psi_B(\theta) . \qquad (9)$$

The following relationship is developed for the second virial coefficient of He as:

$$B_{He}\,[\mathrm{cm}^3/mole] = 8.4 - 0.0018\times T + \frac{115.}{\sqrt{T}} - \frac{835.}{T} , \qquad (10)$$

which agrees with the experimental data of Sengers et al..[32] Similarly, the compiled experimental data of the normalized 3rd virial coefficient for Ne, Ar and Kr are correlated for 1.0 < θ < 45 (Figure 5), as:

$$\left(C/V^{*2}\right) = 0.0757 + \left(-0.0862 - 3.6\times10^{-5}\times\theta + \frac{0.0237}{\theta^{0.059}}\right)\times\tanh\left(0.84\times\theta\right) , \qquad (11)$$

which agrees with > 90% of the experimental data to within ± 5%. For He, 3rd virial coefficient can be neglected.

D. Enthalpy and Specific Heat of the Noble Gases

The thermal capacities and specific heat ratio γ can be calculated from the relationship between pressure, density and temperature (Eq. (8)) using fundamental thermodynamic relationships:[33,34]

$$\hat{H}(T,P) = \hat{H}^o(T_o) + \hat{C}_P^o(T - T_o) + \hat{\rho}\, R_g T\left[\left(B - T\frac{dB}{dT}\right) + \hat{\rho}\left(C - \frac{T}{2}\frac{dC}{dT}\right)\right] \tag{12}$$

$$\begin{aligned}\hat{C}_P = \left[\frac{\partial \hat{H}}{\partial T}\right]_P &= \hat{C}_P^o + \hat{\rho}\, R_g\left[\left(B - T\frac{dB}{dT} - T^2\frac{d^2B}{dT^2}\right) + \hat{\rho}\left(C - \frac{T^2}{2}\frac{d^2C}{dT^2}\right)\right] + \\ &R_g T\left[\left(B - T\frac{dB}{dT}\right) + \hat{\rho}\left(2C - T\frac{dC}{dT}\right)\right] \times \left[\frac{\partial \hat{\rho}}{\partial T}\right]_P,\end{aligned} \tag{13}$$

where

$$\left[\frac{\partial \hat{\rho}}{\partial T}\right]_P = -\frac{\left(\hat{\rho} + B\hat{\rho}^2 + C\hat{\rho}^3\right)/T + \frac{dB}{dT}\hat{\rho}^2 + \frac{dC}{dT}\hat{\rho}^3}{1 + 2B\hat{\rho} + 3C\hat{\rho}^2}\ , \tag{14}$$

and

$$\hat{C}_\upsilon = \left[\frac{\partial \hat{U}}{\partial T}\right]_{\hat{\rho}} = \left[\frac{\partial \hat{H}}{\partial T}\right]_{\hat{\rho}} - \left[\frac{\partial}{\partial T}\left(\frac{P}{\hat{\rho}}\right)\right]_{\hat{\rho}} = \hat{C}_\upsilon^o - \hat{\rho}\, R_g T\left[\left(2\frac{dB}{dT} + T\frac{d^2B}{dT^2}\right) + \hat{\rho}\left(\frac{dC}{dT} + \frac{T}{2}\frac{d^2C}{dT^2}\right)\right], \tag{15}$$

where $\hat{C}_P^o = 5R_g/2$ and $\hat{C}_\upsilon^o = 3R_g/2$ are the ideal values for a dilute gas, and the virial coefficients B and C are functions of temperature only. Calculated values using Eq. (12) are in good agreement with recommended values[30] for He, Ne, Ar, Kr and Xe at up to 20 MPa and temperatures up to 1500, 700, 700, 800 and 800 K, respectively.

IV. Law of Corresponding States for Dense Gas Mixtures

Eqs. (12)–(15) are applicable for predicting the thermodynamic properties of noble gas mixtures, when using the virial coefficients B, C and density $\hat{\rho}$ of the mixtures. This is demonstrated by comparing the predictions of the thermophysical properties for dense noble gas mixtures with the compiled experimental data.

A. Density of Binary Gas Mixtures

Equation (8) applies, whereas the virial coefficients of the mixture are given as:[2,28,36]

$$\overline{B} = x_1^{\,2}B_{11} + 2x_1x_2B_{12} + x_2^{\,2}B_{22}\ , \quad \text{and} \tag{16a}$$

$$\overline{C} = x_1^{\,3}C_{111} + 3x_1^{\,2}x_2C_{112} + 3x_2^{\,2}x_1C_{122} + x_2^{\,3}C_{222}\ . \tag{16b}$$

The virial interaction coefficients are expressed in terms of those of the pure gases, using combination rules as:[35]

$$B_{12} = (B_1 + B_2)/2 + E_{12}\ , \tag{17a}$$

$$C_{112} = \left(C_1^{\,2}C_2\right)^{1/3}, \text{ and } \ C_{122} = \left(C_1C_2^{\,2}\right)^{1/3}. \tag{17b}$$

The defect in the second virial interaction coefficient, E_{12}, is usually a function of temperature and molecular weights of pure gases. The 2nd virial coefficient for the pure noble gases (except helium) is written as:

$$B_{ii} = V_{ii}^* \times \Psi_B(\theta_i), \text{ where } \ V_{ii}^* = R_g\,(T_{cr}/P_{cr})_i\,. \tag{18}$$

According to the principle of corresponding states,[28] the 2nd virial interaction coefficients B_{ij} have a similar form as:

$$B_{ij} = V_{ij}^* \times \Psi_B\left(T/T_{ij}^*\right). \tag{19}$$

A number of mixing rules have been proposed in the literature to express V_{ij}^* and T_{ij}^* in terms of the pure gas values.[28,35-38] The validity of these mixing rules is examined in this paper by comparing their predictions to the compiled experimental data for the B_{12} coefficients for the noble-gas binary mixtures. All mixing rules are valid for the binary mixtures of the heavy noble gases, Ar, Kr and Xe, which have similar critical volumes and temperatures (Table 1). However, most rules compared poorly with the compiled data for binary mixtures of the low molecular weight gases, Ne and He. The mixing rules used in this work are the linear average of the characteristic volumes:[28]

$$V_{ij}^* = (V_{ii}^* + V_{jj}^*)/2\,, \tag{20}$$

and the characteristic temperature, proposed by Prausnitz[28] based on the London theory of dispersion forces, as:

$$T_{ij}^* = \frac{4\beta}{(1+\beta)^2}\sqrt{T_{cr_i}T_{cr_j}}\ , \text{ where } \ \beta = V_{ii}^*/V_{jj}^*\ . \tag{21}$$

Equations (20) and (21) compare well with the compiled experimental data of B_{12} for the binary mixtures Ar-Kr, Ar-Xe and Kr-Xe, and with the data of Ne binary mixtures (Figure 6).

For binary mixtures of He, V_{ij}^* and T_{ij}^* are determined using the values of B_{12} obtained from the fit of Eq. (19) to experimental data. This approach predicts 2nd virial coefficients for He binary mixtures with an error <1.5 cm^3/mole (Figure 6). Note that the molecular weight of the binary gas mixture in Eq. (8) is given by:

$$\overline{M} = x_1 M_1 + x_2 M_2\ . \tag{22}$$

When compared with compiled experimental data of the density of binary mixtures, He-Ne, He-Ar, He-Kr and Ne-Ar near room temperature and up to 12, 23, 3 and 12 MPa, respectively, the predictions of Eq. (8) agree with data to within less than 1%.

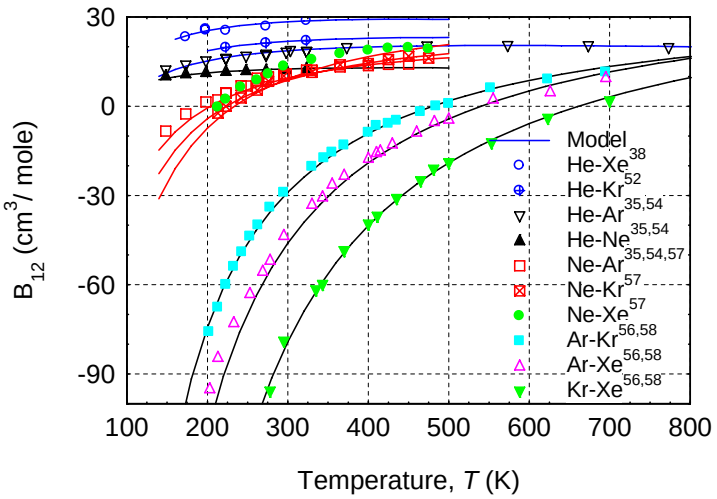


Figure 6. Second virial interaction coefficients.

B. Dynamic Viscosity of Dense Binary Gas Mixtures

Equation (2) is extended to predict the viscosity of binary gas mixtures as:

$$\overline{\mu}(T,P) = \overline{\mu}^o(T) + \left(1 - \frac{1}{2.3}\right)\overline{\mu}^* \times \Psi_\mu\left(\frac{0.291\times\overline{V}^*}{\overline{M}}\rho\right), \tag{23a}$$

where the "pseudo-critical" viscosity of the mixture, $\overline{\mu}^*$, is calculated as:

$$\overline{\mu}^* = 0.204\times10^{-7}\frac{\sqrt{\overline{M}\,\overline{T}^*}}{\left(0.291\times\overline{V}^*\right)^{2/3}}\ . \tag{23b}$$

Equation (23b) makes use of the fact that the compressibility factor of noble gases at the critical point is nearly constant ~ 0.291 (Table 1). The "pseudo-critical" temperature and molar volume of the binary gas mixtures are calculated using the following mixing rules:

$$\overline{V}^* = \sum_{i=1}^{N}\sum_{j=1}^{N} x_i x_j V_{ij}^* \quad \text{and} \quad \overline{V}^*\overline{T}^* = \sum_{i=1}^{N} x_i V_{ii}^* T_{ii}^*\ . \tag{24}$$

The parameters V_{ij}^* and T_{ij}^* are given by Eqs. (20) and (21). Equation (24) successfully predicts the critical temperatures of methane-nitrogen mixtures.[39] The van der Waals' mixing rule, Eq. (24a), reduces to:

$$\overline{V}^* = x_1{}^2 V_{11}^* + 2x_1x_2\left(\frac{V_{11}^* + V_{22}^*}{2}\right) + x_2{}^2 V_{22}{}^* = x_1 V_{11}^* + x_2 V_{22}^*\ . \tag{25}$$

The present work examined various mixing rules[10,36,40] and found that the predictions of Eqs. (24a) and (24b) are the closest to the compiled experimental data for the viscosity of the noble gas mixtures to within less than 1.5%. The predictions are compared with the viscosity measurements of Ne-Ar gas mixtures in Figure 7. Also shown in this figure are the predictions when Eq. (24b) is replaced with the second mixing rule of van der Waals:[40]

$$\overline{V}^*\overline{T}^* = \sum_{i=1}^{N}\sum_{j=1}^{N} x_i x_j V_{ij}^* T_{ij}^*\ . \tag{26}$$

For binary gas mixtures involving He, Eq. (23a) is modified to account for the fact that the viscosity of helium is independent of density. The linear rule is used to evaluate the defect viscosity of the mixture at the critical point, and assuming that the defect viscosity of He is nil, gives:

$$\overline{\Delta\mu}^{*} = x_1 \Delta\mu_{cr1} + x_2 \Delta\mu_{cr2} \approx x_1 \Delta\mu_{cr1} \ . \quad (27)$$

The subscript '2' represents helium and the subscript '1' refers to the heavier gas in the mixture. Equation (23a) is then reduced to the following expression for He binary gas mixtures:

$$\overline{\mu}(T,P) = \overline{\mu}^{o}(T) + 0.565\, x_1\, \mu_1^{*}\, \Psi_{\mu}\left(\frac{0.291 \times \overline{V}^{*}}{\overline{M}}\, \rho \right) . \quad (28)$$

Equation (28) predicts the compiled data for the viscosity of He-Ne, He-Ar and He-Kr binary gas mixtures to within 1.5%, at pressures up to 12.2 MPa and reduced densities up to 0.23 (for He-Ne). In Eqs. (23a) and (28), $\overline{\mu}^{o}(T)$ is the viscosity of a dilute binary gas mixture, calculated using the Chapman-Enskog kinetic approach as:[3]

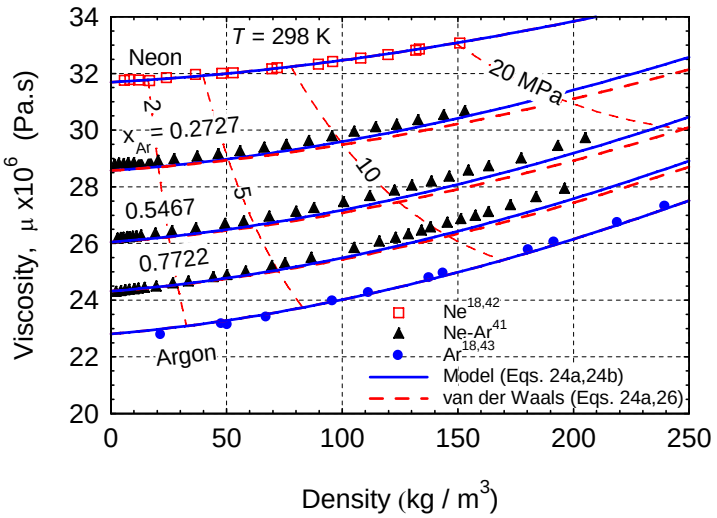


Figure 7. Viscosity of Ne-Ar mixtures at 289 K.

$$\overline{\mu}^{o} = \left[\frac{x_1^{\,2}}{H_{11}} - \frac{2 x_1 x_2 H_{12}}{H_{22} H_{11}} + \frac{x_2^{\,2}}{H_{22}} \right] \times \left[1 - \frac{H_{12}^{\,2}}{H_{22} H_{11}} \right]^{-1} . \quad (29)$$

The interaction coefficients $H_{12} << \sqrt{H_{11} H_{22}}$ for all noble-gas binary mixtures are negligible, reducing Eq. (29) to the form proposed by Sutherland and Wassiljewa nearly a century ago:

$$\overline{\mu}^{o} = \mu_1^{o} \Big/ \left(1 + \phi_{12}\, x_2 / x_1\right) + \mu_2^{o} \Big/ \left(1 + \phi_{21}\, x_1 / x_2\right) , \quad (30)$$

where the interaction coefficients ϕ_{ij} are given by:[3]

$$\phi_{12} = \frac{\mu_1^{o}}{\mu_{12}} \left[\frac{2 m_1 m_2}{(m_1 + m_2)^2} \right] \times \left[\frac{5}{3 A_{12}^{*}} + \frac{m_2}{m_1} \right], \quad \phi_{21} = \frac{\mu_2^{o}}{\mu_{12}} \left[\frac{2 m_1 m_2}{(m_1 + m_2)^2} \right] \times \left[\frac{5}{3 A_{12}^{*}} + \frac{m_1}{m_2} \right] . \quad (31)$$

In these equations, $A_{12}^{*} = \Omega^{(2,2)*} / \Omega^{(1,1)*}$, a very slow varying function of $T_{12}^{*} = kT / \varepsilon_{12}$, is taken constant (1.10). For the Lennard-Jones (6,12) potential,[2] 1.094 < A_{12}^{*} < 1.119 in the temperature range 1.0 < T_{12}^{*} < 20. Based on the compiled experimental data of the viscosity of dilute noble-gas binary mixtures at different temperatures, the following expression for the interaction viscosities is obtained:

$$\mu_{12} = A_{\mu} (T - T_{\mu})^{n} . \quad (32)$$

The coefficients A_{μ} and T_{μ} are listed in Table 2, and the values of exponent n are given in Table 3.

Table 2. Coefficients A_{μ} and T_{μ} for the interaction viscosity μ_{12} of noble gas mixtures, Eq. (32).

		$A_{\mu} \times 10^{7}$				
		Helium	Neon	Argon	Krypton	Xenon
T_{μ} (K)	Helium		8.81837	6.57562	11.2472	3.40998
	Neon	63.63		10.4450	15.2944	18.2681
	Argon	51.87	47.66		18.5326	16.9684
	Krypton	121.27	94.86	148.93		11.2303
	Xenon	45.89	125.0	151.58	116.71	

The predictions of Equations (30)-(32) are in excellent agreement with the compiled experimental data of binary mixtures of Ne, Ar, Kr and Xe to within ± 2.5%, and to within ± 4% of the data for He binary mixtures at temperatures up to 1533 K.[44,45] These equations are also in good agreement with the calculated values using the Chapman-Enskog approach.[4]

Table 3. Exponent *n* Eq. (32), and correction factor f_{12}, Eq. (36).

		Exponent *n*				
		Helium	Neon	Argon	Krypton	Xenon
f_{12}	Helium		0.614098	0.602712	0.508158	0.658754
	Neon	0.895		0.620956	0.565158	0.522034
	Argon	0.940	0.845		0.541593	0.542416
	Krypton	1.020	0.900	0.850		0.631571
	Xenon	1.060	0.940	0.935	0.871	

C. Thermal Conductivity of Dense Binary Gas Mixtures

Equation (6) for pure gases is extended to predict the conductivity of binary gas mixtures as:

$$\overline{\lambda}(T,P) = \overline{\lambda}^{o}(T) + \left(1 - \frac{1}{2.94}\right)\overline{\lambda}^{*} \times \Psi_{\lambda}\left(\frac{0.291 \times \overline{V}^{*}}{\overline{M}}\rho\right). \tag{33a}$$

The "pseudo-critical" conductivity of the mixture, $\overline{\lambda}^{*}$, is calculated as:

$$\overline{\lambda}^{*} = 0.304 \times 10^{-4}\,(\overline{T}^{*})^{0.277} \Big/ \left(\overline{M}^{0.465}\left(0.291 \times \overline{V}^{*}\right)^{0.415}\right). \tag{33b}$$

The "pseudo-critical" temperature and volume of the binary mixtures are calculated using the van der Waals' mixing rules, Eqs. (24a) and (26). Among the various mixing rules reported in the literature[10,36,46] and examined in this work, these equations are the most accurate when compared with the compiled experimental data for the thermal conductivity of He-Ne, He-Ar and Ne-Ar gas mixtures,[53,54] with a maximum deviation of 5.0%. In Eq. (33a), $\overline{\lambda}^{o}(T)$ is calculated using the Chapman-Enskog approach as:[3]

$$\overline{\lambda}^{o} = \left[\frac{x_1^{\,2}}{L_{11}} - \frac{2x_1x_2L_{12}}{L_{22}L_{11}} + \frac{x_2^{\,2}}{L_{22}}\right] \times \left[1 - \frac{L_{12}^{\,2}}{L_{22}L_{11}}\right]^{-1} \tag{34}$$

where

$$L_{11} = \frac{x_1^{\,2}}{\lambda_1^{o}} + \frac{x_1x_2}{2\lambda_{12}} \times \frac{(15/2)m_1^{\,2} + (25/4)m_2^{\,2} - 3m_2^{\,2}B_{12}^{*} + 4m_1m_2A_{12}^{*}}{(m_1+m_2)^2 A_{12}^{*}}\ , \tag{35a}$$

$$L_{22} = \frac{x_2^{\,2}}{\lambda_2^{o}} + \frac{x_1x_2}{2\lambda_{12}} \times \frac{(15/2)m_2^{\,2} + (25/4)m_1^{\,2} - 3m_1^{\,2}B_{12}^{*} + 4m_1m_2A_{12}^{*}}{(m_1+m_2)^2 A_{12}^{*}}\ , \tag{35b}$$

$$L_{12} = -\frac{x_1x_2}{2\lambda_{12}} \times \frac{m_1m_2}{(m_1+m_2)^2 A_{12}^{*}} \times \left(55/4 - 3B_{12}^{*} - 4A_{12}^{*}\right). \tag{35c}$$

The coefficient $B_{12}^{*} = [5\Omega^{(1,2)*} - 4\Omega^{(1,3)*}]/\Omega^{(1,1)*}$, a slow varying function of $T_{12}^{*} = kT/\varepsilon_{12}$, is taken constant (1.10). For the Lennard-Jones (6,12) potential,[2] $1.090 < B_{12}^{*} < 1.119$ for $2.0 < T_{12}^{*} < 20$; and λ_{12} is related to μ_{12} as:

$$\lambda_{12} = \frac{15}{4}\frac{k}{m_{12}}\mu_{12}f_{12}\ , \tag{36}$$

Where $m_{12} = 2m_1m_2/(m_1+m_2)$. The correction factor f_{12} compensates for the error involved in using only a first-order kinetic-theory formula for $\overline{\lambda}^{o}$. A constant f_{12} is selected for each mixture, based on compiled experimental data (Table 3). The present approach predicts ~ 87% of the low-density conductivity data of noble gas binary mixtures to within $\pm$ 5% and is in excellent agreement with the predictions of Chapman-Enskog for dilute gases.[4] This is reasonably good, considering the large scattering in thermal conductivity compared to the viscosity data.

V. Effects of Pressure

The property correlations developed in the previous sections are used to evaluate the effects of pressure on the properties of pure noble gases and their binary mixtures. The presented results at typical operating conditions at the exit of the compressor in a space nuclear reactor power system with CBC engines, a system pressure of 2.0 MPa and 400 K, are compared with those at 0.1 MPa. The noble gases and their binary mixtures behave essentially like perfect gases at $\geq$ 400 K and atmospheric pressure (0.1 MPa), for which *Z* is unity. At 2.0 MPa, *Z* is slightly lower,

0.987 and 0.957 for Kr and Xe, and for binary mixtures of molecular weights M > 70. At > 2.0 MPa, the compressibility factors for Ar, Kr and Xe and binary gas mixtures with M > 40 deviate further from unity and thus, the virial Equation of state (8) is used.

The molar heat capacity at constant pressure of noble gases and their binary mixtures at 400 K and atmospheric pressure is nearly equal to the ideal value, 20.786 J/mole.K. Larger deviations occur at higher pressure of 2.0 MPa, and increase as the molecular weight increases. At 2.0 MPa and 400 K, the molar heat capacities of Ar, Kr and Xe (21.257, 21.72 and 23.00 J/mole.K, respectively) are 2.3%, 4.5% and 10.7% higher than the ideal gas value. For all binary mixtures with molecular weights < 40, the molar heat capacity is within 2.3% of that of a perfect gas. Much larger deviations occur at pressures > 2.0 MPa, particularly for noble gases of high molecular weight and their binary mixtures. The specific heat ratio for noble gases and their binary mixtures is nearly equal to the ideal value of 5/3 at atmospheric pressure. For the heavier gases, large deviations occur at 2.0 MPa, where γ = 1.697, 1.724 and 1.786 for Ar, Kr and Xe, which correspond to 1.8%, 3.4% and 7.1% deviations from the ideal value of 5/3.

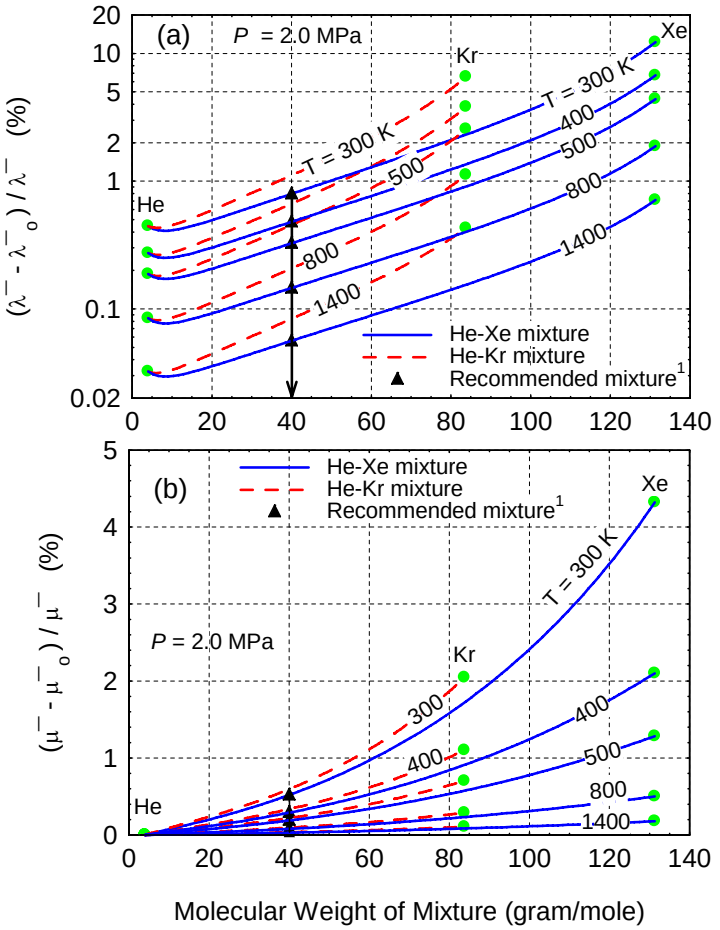


Figure 8. Dynamic viscosities and thermal conductivities of dense binary gase mixtures.

Figures 8a and 8b shows the errors of neglecting the effect of pressure on the thermal conductivity and dynamic viscosity of He, Kr, Xe and their binary mixtures, which increase with increasing the gas density (i.e. decreasing temperature) and / or the molecular weight. Figure 8a shows that the difference between the thermal conductivity of a dense He-Xe mixture at 400 K and 2.0 MPa and that at 0.1 MPa increases steadily with increasing Xe. For the same molecular weight, the differences in the thermal conductivities of dense and the dilute He-Kr mixtures are higher than those for the He-Xe mixtures. This is because the thermal conductivity of a dilute the He-Kr (at 0.1 MPa) is lower than that of a He-Xe, of the same molecular weight, while the thermal conductivity defect at higher pressure is nearly identical for the two mixtures. At 2.0 MPa and > 300 K, the error in the thermal conductivity at 2.0 MPa, relative to that at 0.1 MPa, is < 1% for He-Kr and for pure Ar it is about 5 times that for the two helium binary gas mixtures, of the same molecular weight of 40 (Fig. 8a). For example, at 300 K and 2.0 MPa, neglecting the effect of pressure results in 4% error in the thermal conductivity of Ar, compared to only about 1% for He-Xe and He-Kr binary mixtures, of the same molecular weight. At atmospheric pressure and T > 300 K, the errors of neglecting the effect of pressure on the thermal conductivities of Ar and theHe-Kr and He-Xe binary mixtures is < 0.2% validating the applicability of the Chapman-Enskog approach for dilute gases at these conditions.

The errors of neglecting the effect of pressure on the dynamic viscosities of noble gases and their binary mixtures are smaller than on the thermal conductivities (Figure 8b). One reason is the peculiar behavior of helium, whose dynamic viscosity is independent of pressure.[11,18,19] For example, at 300 K and 2.0 MPa, the viscosity error for a He_{72}-Xe_{28} (or molecular weight of 40) is 0.5% (Figure 8b), compared to 0.8% in the thermal conductivity (Figure 8a). For Ar at the same conditions, the error of neglecting the effect of pressure on the thermal conductivity and dynamic viscosity is higher, ~4% and ~1%. At 2.0 MPa, and 300 K, the error in the thermal conductivity and viscosity Xe due to neglecting the effect of pressure is 12.3% and 4.3% (Figures 8a and 8b).

In summary, the effect of pressure on the transport properties of the $He_{72}Xe_{28}$ binary mixture, recommended working fluid for space nuclear reactor power systems with CBC engines[1], is negligible, and much less pronounced than for pure Ar. Helium behaves essentially like a perfect dilute gas, while xenon exhibits the largest effect of

pressure on transport properties (Figures 8a and 8b), because of its much higher critical temperature and density (Table 1).

VI. Summary and Conclusions

Experimental data of thermodynamic and transport properties of the noble gases and their binary mixtures, for temperatures up to 1400 K and pressures up to 20 MPa, are compiled and used to quantify the effect of increasing the gas density on these properties. The properties of interest include density, enthalpy, specific heats, the dynamic viscosity, and the thermal conductivity of the 5 noble gases and their 10 binary mixtures. The semi-empirical models of these properties developed herein, which are similar to those for dilute gases, employ the two-parameter principle of corresponding states to account for the effect of pressure. The expressions developed are valid at temperatures between the critical value, T_{cr} and 1400 K, pressures up to 20 MPa, and densities up to the critical density, ρ_{cr}.

The errors in the properties, due to neglecting the effect of pressure, are computed and compared as functions of temperature, for dense noble gases and their binary mixtures, particularly He-Xe with a molecular weight of 40 gram/mole (72 mole% helium and 28 mole% xenon). The latter is the recommended binary mixture for use in space nuclear reactor power systems with CBC engines, at 400 K to 1400 K and ~ 2.0 MPa. Results show that at these operating conditions, He, Ne, and the He-Xe and He-Kr binary gas mixtures with a molecular weight of 40 gram/mole behave like a perfect gas and the errors of neglecting the effect of pressure on their compressibility factor, specific heats, and transport properties are 1%.

Acknowledgments

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